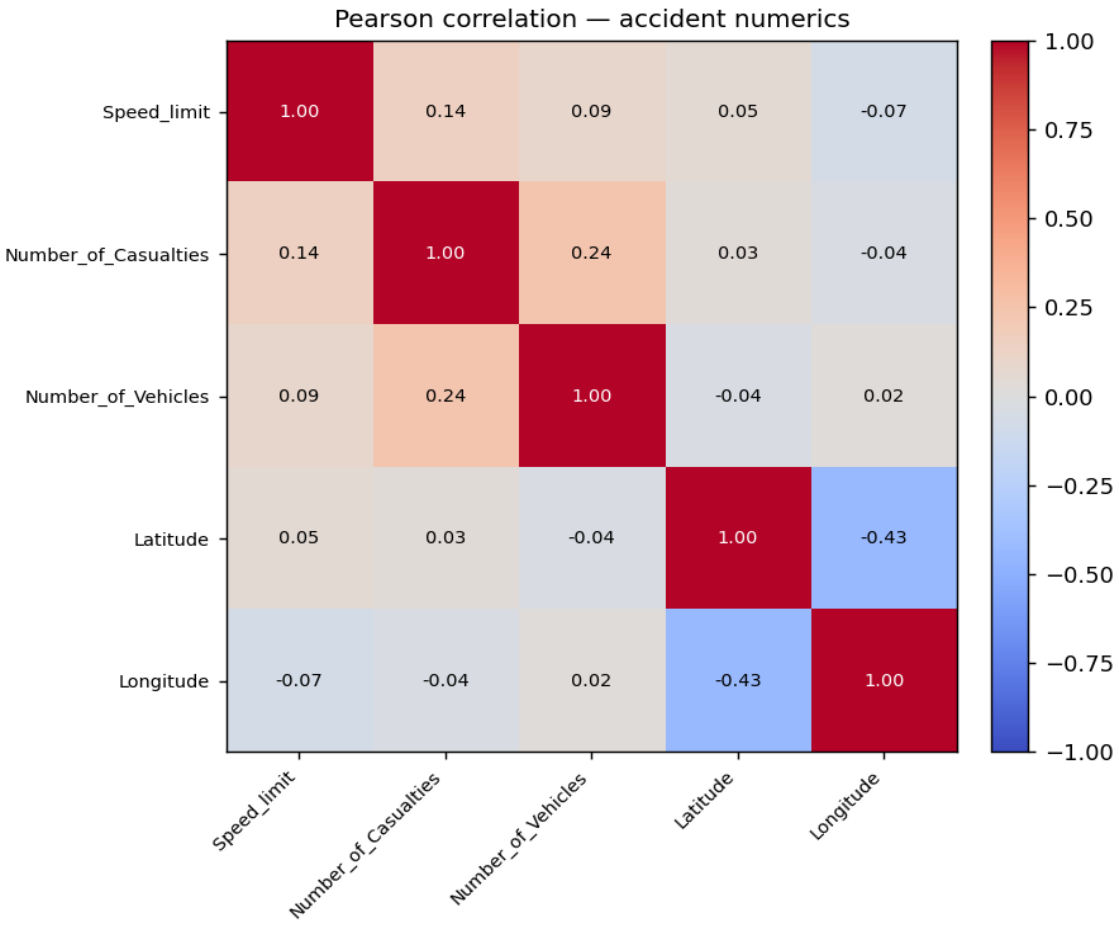
UK Road Accidents — Data Validation Report

Generated: 2026-04-18T15:54:59.538258

Total Issues Found: 14

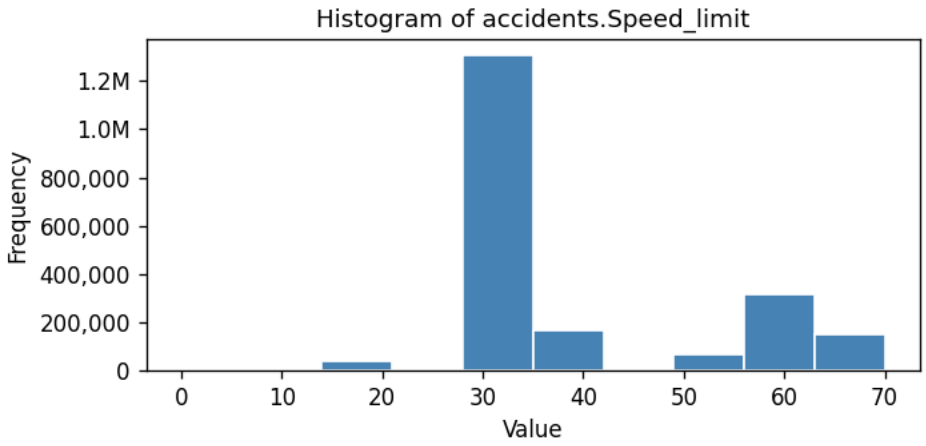
Dimension	Issues
ACCURACY	2
COMPLETENESS	10
JOIN_INTEGRITY	2

Correlation Heatmap

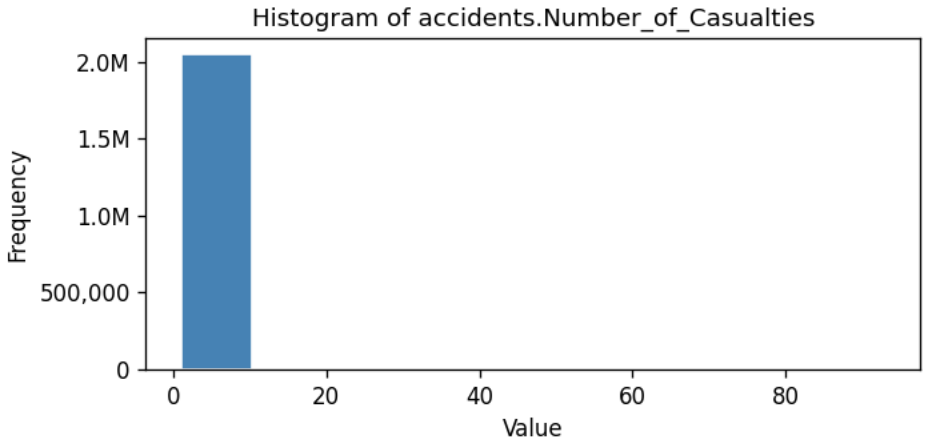


Distribution Histograms

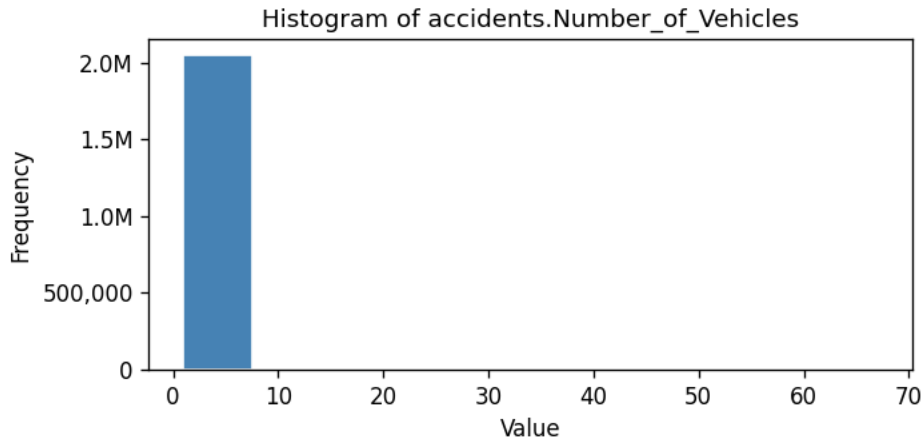
accidents.Speed_limit



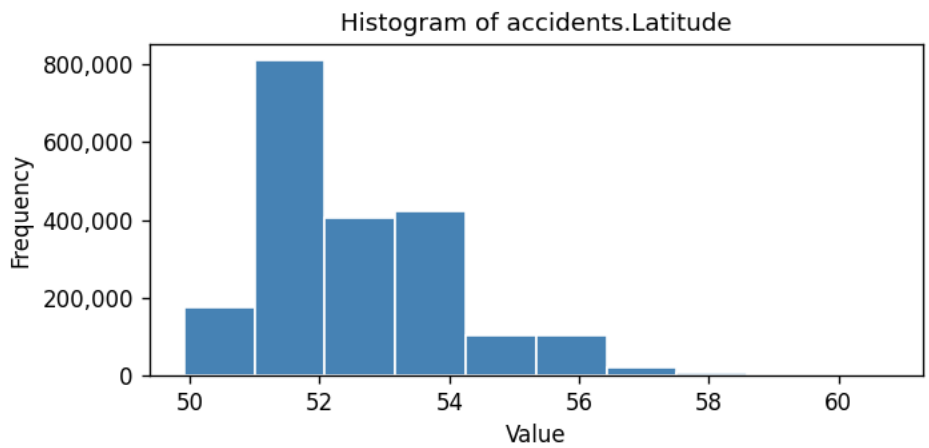
accidents.Number_of_Casualties



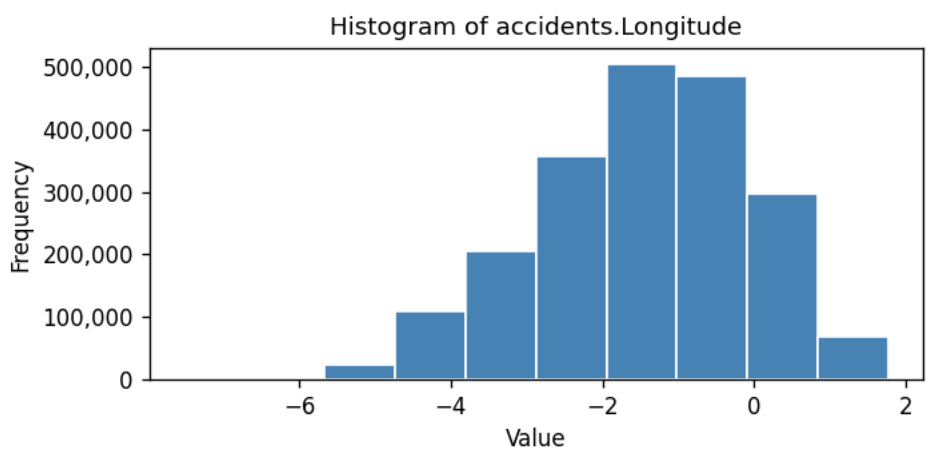
accidents.Number_of_Vehicles



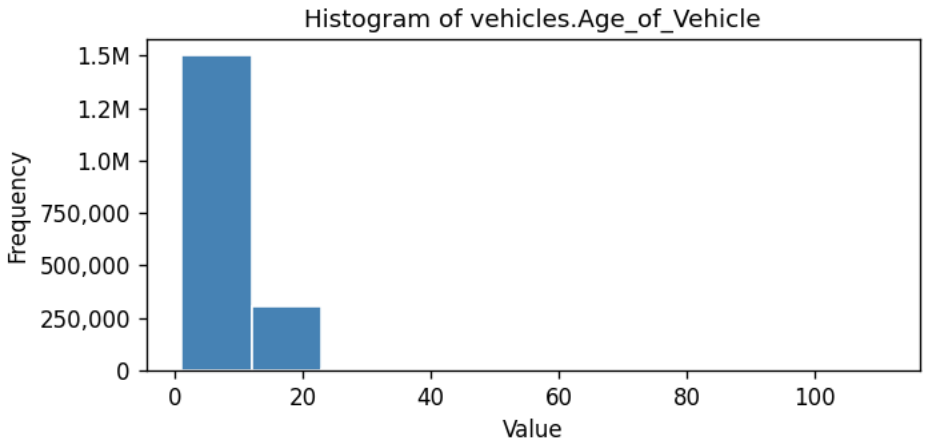
accidents.Latitude



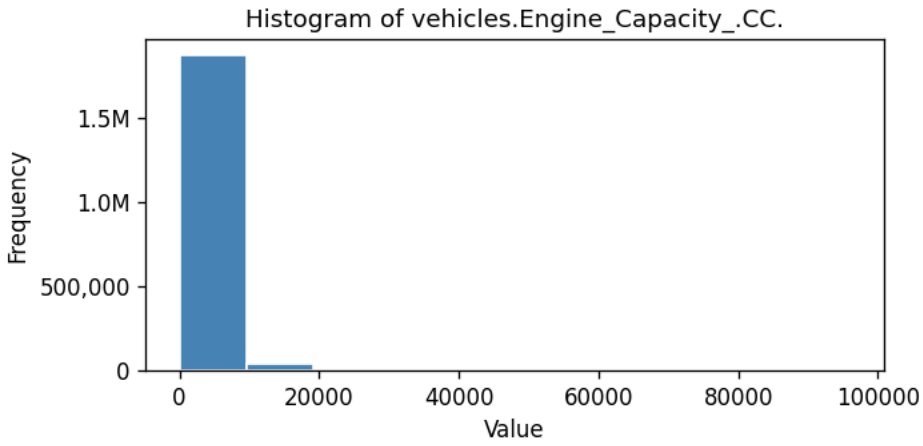
accidents.Longitude



vehicles.Age_of_Vehicle



vehicles.Engine_Capacity_.CC.



Full Validation Log

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================================================================
UK Road Accidents — Data Validation Report
2026-04-18 15:54:59
================================================================
Loaded accidents : 2,047,256 rows x 34 columns
Loaded vehicles : 2,177,205 rows x 24 columns
----------------------------------------------------------------
DIMENSION 1 — ACCURACY
----------------------------------------------------------------
[PASS] Accident_Severity — all values valid (Slight/Serious/Fatal)
[ISSUE] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[PASS] Latitude — all within UK bounds (49–61)
[PASS] Longitude — all within UK bounds (-8 to 2)
[PASS] Number_of_Vehicles — all values >= 1
[PASS] Age_of_Vehicle — no negative values
[INFO] Age_Band_of_Driver has 12 unique categories
Sample values: ['26 - 35', '66 - 75', '36 - 45', '46 - 55', '21 - 25', 'Data missing or out of range', '16 - 20', '
56 - 65', 'Over 75', '11 - 15']
[PASS] Day_of_Week — all values valid
[ISSUE] 76,119 vehicle records have invalid Sex_of_Driver values
----------------------------------------------------------------
DIMENSION 2 — CONSISTENCY
----------------------------------------------------------------
[PASS] Date column — parses correctly as datetime
[PASS] Year column — matches year extracted from Date in all rows
[WARN] 2 Date/Day_of_Week mismatches found
[PASS] Time column — all values follow HH:MM format
[PASS] Year — consistent across both tables for all matched accidents
----------------------------------------------------------------
DIMENSION 3 — COMPLETENESS
----------------------------------------------------------------
[ACCIDENTS]
Column Missing %
------------------------------------------- --------- ------
Carriageway_Hazards 2,007,807 98.1%
Special_Conditions_at_Site 1,995,137 97.5%
2nd_Road_Class 844,272 41.2%
LSOA_of_Accident_Location 144,953 7.1%
2nd_Road_Number 17,593 0.9%
Pedestrian_Crossing-Physical_Facilities 3,560 0.2%
Pedestrian_Crossing-Human_Control 2,920 0.1%
Did_Police_Officer_Attend_Scene_of_Accident 278 0.0%
Latitude 174 0.0% <- REQUIRED
Location_Easting_OSGR 164 0.0%
Location_Northing_OSGR 164 0.0%
Longitude 175 0.0% <- REQUIRED
... and 4 more columns with missing values
[ISSUE] [ACCIDENTS] Required column 'Latitude' has 174 null values (0.0%)
[ISSUE] [ACCIDENTS] Required column 'Longitude' has 175 null values (0.0%)
[ISSUE] [ACCIDENTS] Required column 'Speed_limit' has 37 null values (0.0%)
[ISSUE] [ACCIDENTS] 'Carriageway_Hazards' is 98.1% missing (>20.0% threshold)
[ISSUE] [ACCIDENTS] 'Special_Conditions_at_Site' is 97.5% missing (>20.0% threshold)
[ISSUE] [ACCIDENTS] '2nd_Road_Class' is 41.2% missing (>20.0% threshold)
[VEHICLES]
Column Missing %
------------------------------------------- --------- ------
Hit_Object_in_Carriageway 2,087,824 95.9%
Hit_Object_off_Carriageway 1,989,741 91.4%
Skidding_and_Overturning 1,898,208 87.2%
Driver_IMD_Decile 734,812 33.8%
Age_of_Vehicle 358,149 16.4%
model 325,331 14.9%
Engine_Capacity_.CC. 265,861 12.2%
Propulsion_Code 245,843 11.3%
make 110,845 5.1%
Vehicle_Location.Restricted_Lane 1,317 0.1%
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[ISSUE] [VEHICLES] 'Hit_Object_in_Carriageway' is 95.9% missing (>20.0% threshold)
[ISSUE] [VEHICLES] 'Hit_Object_off_Carriageway' is 91.4% missing (>20.0% threshold)
[ISSUE] [VEHICLES] 'Skidding_and_Overturning' is 87.2% missing (>20.0% threshold)
[ISSUE] [VEHICLES] 'Driver_IMD_Decile' is 33.8% missing (>20.0% threshold)
----------------------------------------------------------------
DIMENSION 4 — UNIQUENESS
----------------------------------------------------------------
[PASS] Accidents — zero fully duplicate rows
[PASS] Accidents — all 2,047,256 Accident_Index values are unique
[PASS] Vehicles — zero fully duplicate rows
[INFO] Vehicles — 688,224 duplicate Accident_Index values (expected by design)
[PASS] Vehicles — (Accident_Index, Vehicle_Reference) composite key is fully unique
----------------------------------------------------------------
DIMENSION 5 — OUTLIERS
----------------------------------------------------------------
IQR Method (bounds = Q1 - 1.5*IQR and Q3 + 1.5*IQR):
Column Outliers Lower fence Upper fence
------------------------------------------ ---------- ------------ ------------
Isolation Forest (contamination=0.05, multivariate):
accidents: 4,858 anomalies in 100,000 rows (4.9%)
vehicles: 4,999 anomalies in 100,000 rows (5.0%)
[NOTE] Outliers are FLAGGED only — removal decisions made in cleaning step.
----------------------------------------------------------------
DIMENSION 6 — TIMELINESS
----------------------------------------------------------------
Earliest record : 2005-01-01
Latest record : 2017-12-31
[PASS] No future dates
[PASS] No dates before 2005
----------------------------------------------------------------
DIMENSION 7 — DISTRIBUTION PROFILE
----------------------------------------------------------------
Column Min Max Mean Median Std Skew Kurt Card
---------------------------------------- ------- -------- -------- -------- -------- ------- ------- ------
accidents.Speed_limit 0.0 70.0 38.84 30.00 14.15 1.10 -0.36 9
Histogram (accidents.Speed_limit):
[ 0.0 – 7.0] 1 ( 0.0%) ■
[ 7.0 – 14.0] 19 ( 0.0%) ■
[ 14.0 – 21.0] 38,415 ( 1.9%) ■
[ 21.0 – 28.0] 0 ( 0.0%) ■
[ 28.0 – 35.0] 1,306,174 ( 63.8%) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
[ 35.0 – 42.0] 168,357 ( 8.2%) ■■■■
[ 42.0 – 49.0] 0 ( 0.0%) ■
[ 49.0 – 56.0] 69,479 ( 3.4%) ■
[ 56.0 – 63.0] 317,469 ( 15.5%) ■■■■■■■
[ 63.0 – 70.0] 147,305 ( 7.2%) ■■■
accidents.Number_of_Casualties 1.0 93.0 1.35 1.00 0.82 8.57 412.75 52
Histogram (accidents.Number_of_Casualties):
[ 1.0 – 10.2] 2,046,795 (100.0%) ■■■■■■■■■■■■■■■■■■■
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accidents.Latitude 49.9 60.8 52.56 52.24 1.45 1.03 0.83 1,319,367
Histogram (accidents.Latitude):
[ 49.91 – 51.0] 175,171 ( 8.6%) ■■■■
[ 51.0 – 52.08] 809,444 ( 39.5%) ■■■■■■■■■■■■■■■■■■■■
[ 52.08 – 53.17] 405,212 ( 19.8%) ■■■■■■■■■■
[ 53.17 – 54.25] 424,012 ( 20.7%) ■■■■■■■■■■■
[ 54.25 – 55.34] 102,546 ( 5.0%) ■■
[ 55.34 – 56.42] 102,828 ( 5.0%) ■■
[ 56.42 – 57.5] 20,906 ( 1.0%) ■
[ 57.5 – 58.59] 6,073 ( 0.3%) ■
[ 58.59 – 59.67] 492 ( 0.0%) ■
[ 59.67 – 60.76] 398 ( 0.0%) ■
accidents.Longitude -7.5 1.8 -1.41 -1.36 1.40 -0.39 -0.35 1,417,728
Histogram (accidents.Longitude):
[ -7.52 – -6.59] 225 ( 0.0%) ■
[ -6.59 – -5.66] 1,071 ( 0.1%) ■
[ -5.66 – -4.73] 23,534 ( 1.1%) ■
[ -4.73 – -3.8] 108,905 ( 5.3%) ■■
[ -3.8 – -2.88] 204,273 ( 10.0%) ■■■■■
[ -2.88 – -1.95] 356,689 ( 17.4%) ■■■■■■■■
[ -1.95 – -1.02] 503,647 ( 24.6%) ■■■■■■■■■■■■
[ -1.02 – -0.09] 484,383 ( 23.7%) ■■■■■■■■■■■
[ -0.09 – 0.83] 296,607 ( 14.5%) ■■■■■■■
[ 0.83 – 1.76] 67,747 ( 3.3%) ■
vehicles.Age_of_Vehicle 1.0 111.0 7.11 7.00 4.73 1.22 5.72 88
Histogram (vehicles.Age_of_Vehicle):
[ 1.0 – 12.0] 1,500,285 ( 82.5%) ■■■■■■■■■■■■■■■■■■■
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Speed_limit x Number_of_Casualties 0.143 0.162 0.019
Speed_limit x Number_of_Vehicles 0.088 0.069 0.019
Speed_limit x Latitude 0.052 0.034 0.018
Speed_limit x Longitude -0.066 -0.062 0.004
Number_of_Casualties x Number_of_Vehicles 0.237 0.230 0.007
Number_of_Casualties x Latitude 0.030 0.044 0.014
Number_of_Casualties x Longitude -0.038 -0.046 0.008
Number_of_Vehicles x Latitude -0.039 -0.029 0.010
Number_of_Vehicles x Longitude 0.019 0.018 0.001
Latitude x Longitude -0.434 -0.411 0.023
[PASS] No high-correlation pairs found (threshold |r| > 0.85)
[SAVED] reports\correlation_heatmap.png
----------------------------------------------------------------
JOIN INTEGRITY (Accident_Index key across both tables)
----------------------------------------------------------------
Unique Accident_Index in accidents table : 2,047,256
Unique Accident_Index in vehicles table : 1,488,981
Keys matched (will survive inner join) : 1,389,724
[ISSUE] 657,532 accidents (32.1%) have no vehicle record — dropped on inner join, document in cleaning step
[ISSUE] 99,257 vehicle records (6.7%) have no matching accident — orphaned records, dropped on inner join
================================================================
VALIDATION COMPLETE
Total issues found across all 8 dimensions: 14
================================================================
Issues by dimension:
[ACCURACY] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[ACCURACY] 76,119 vehicle records have invalid Sex_of_Driver values
[COMPLETENESS] [ACCIDENTS] Required column 'Latitude' has 174 null values (0.0%)
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[JOIN_INTEGRITY] 657,532 accidents (32.1%) have no vehicle record — dropped on inner join, document in cleaning step
[JOIN_INTEGRITY] 99,257 vehicle records (6.7%) have no matching accident — orphaned records, dropped on inner join
[SAVED] reports\validation_report.json
[SAVED] reports\validation_summary.txt
```